

Module 8: Trees

Part 2: Bagging and Random Forest

Wenting Xu

Iowa State University

Copyright and Attribution

Some lecture slides and instructional materials in this course are adapted from the following sources:

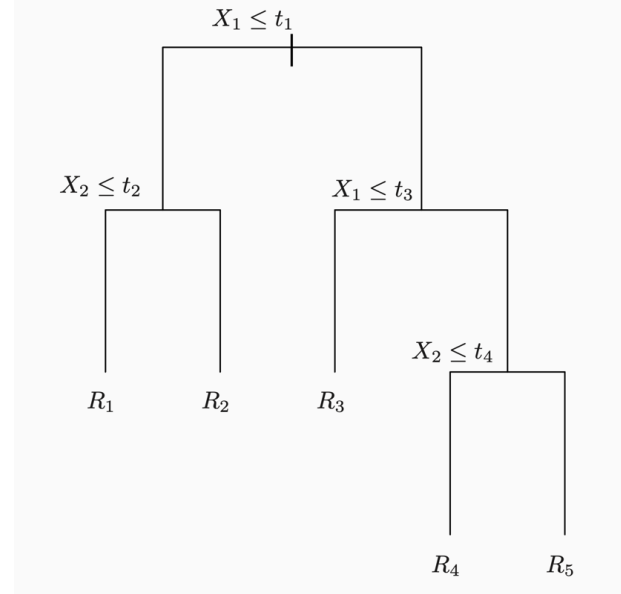
- *An Introduction to Statistical Learning: With Applications in R (Second Edition)*

Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani
Springer, 2021

- Online course materials developed by Trevor Hastie, Robert Tibshirani, and collaborators.

not robust

- Decision trees have **high variance**
→ small data changes → very different trees
- Goal: **reduce variance**



How can we reduce the variance ?

Stats Review

Suppose we are given a set of n **independent** variables

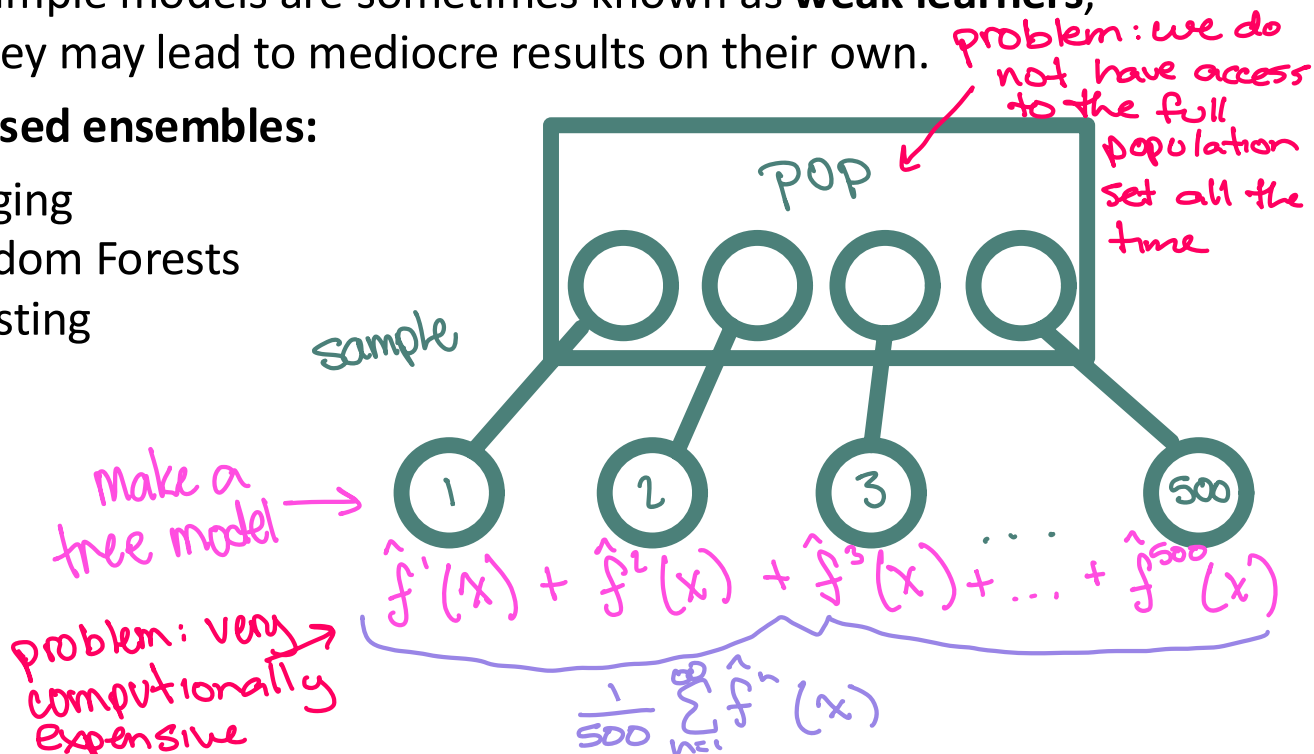
$Z_1, \dots, Z_n$. Each has their own variance σ^2 .

- Variance of $Z_i = \sigma^2$
- Variance of $\bar{Z} = \sigma^2/n \rightarrow \bar{Z} = \frac{Z_1 + Z_2 + \dots + Z_n}{n} = \frac{\sigma^2}{n}$

Averaging a set of observations reduces variance!

Ensemble Methods

- An **ensemble method** is an approach to **combines many simple models** in order to obtain a single and potentially very powerful model.
- These simple models are sometimes known as **weak learners**, since they may lead to mediocre results on their own.
- **Tree-based ensembles:**
 - Bagging
 - Random Forests
 - Boosting



Apply this logic to trees

- How to reduce variance in trees?
- Take the average of them!
- Idea: take **many** training sets from the **population**, construct a tree using each training set, and **average** the resulting predictions:

$$\hat{f}_{\text{bag}(x)} = \frac{1}{B} \sum_{b=1}^B \hat{f}^b(x)$$

Obvious problem with this approach?

Bootstrap

- Generate B different bootstrapped training sets.
- For the b th bootstrapped training set, we get a bootstrap sample and its predictions $\hat{f}^{(b)}(x)$.
- Average all the predictions to obtain:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^b(x)$$

- This is called bagging.

Bootstrap:
Sampling with
replacement

→ construct and
fit a tree
(we will have
 B trees)

Bagging

To apply bagging to regression trees:

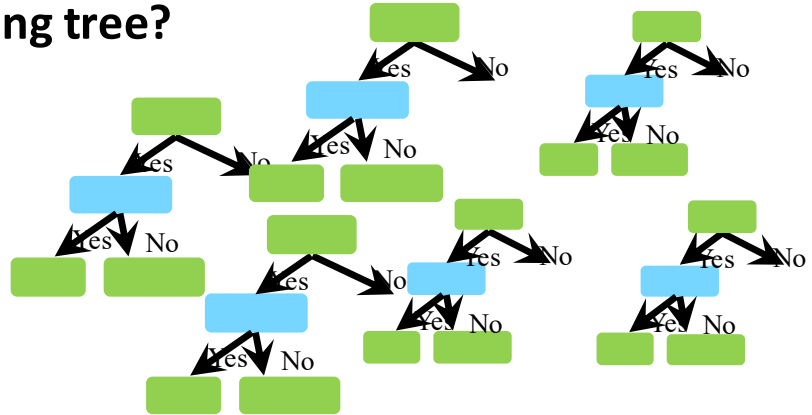
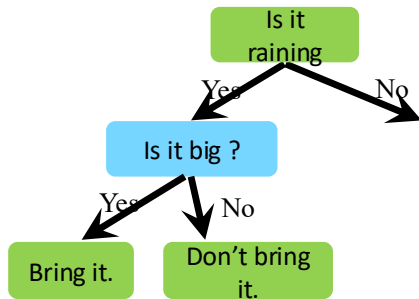
Y = continuous (e.g. salary, price)

- Construct **B regression trees** using **B bootstrapped training sets**, and average the resulting predictions.
- Each individual tree grown deep and **not pruned**.
- Each tree has high variance, but low bias.
- **Averaging these B trees reduces the variance.**
- Improvements in accuracy by combining together hundreds or even thousands of trees into a single procedure.
- Number of B is not critical with bagging. Use a value of B sufficiently large that the error has settled down.

Bagging

Bagging improves prediction accuracy at the expense of interpretability.

How do we interpret the resulting tree?



- Overall summary of the **importance** of each prediction using the RSS (regression trees) or Gini index (classification trees).
- **Regression trees**: record the total amount that the **RSS** is decreased due to splits over a given predictor, averaged over all B trees. A large value indicates an important predictor.
- **Classification trees**: record the total amount the **Gini index** is decreased by splits over a given predictor, averaged over all B trees.

Bagging

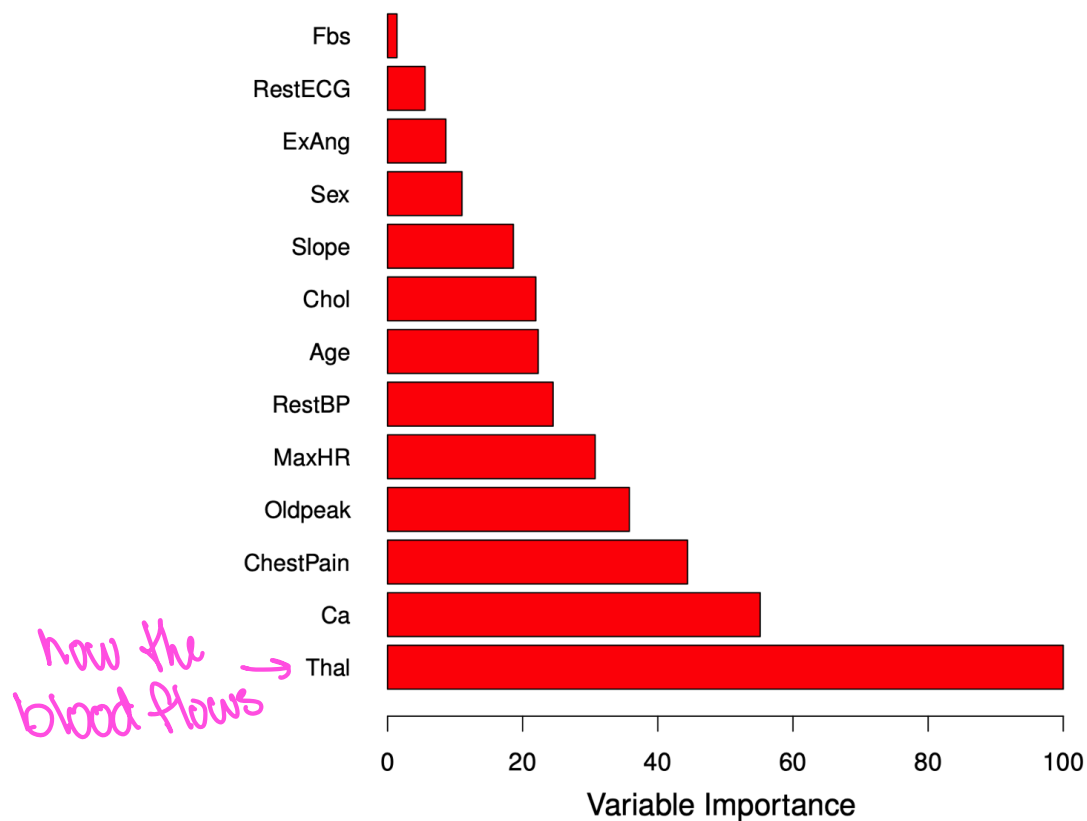


Fig 1. A variable importance plot for the **Heart** data.

Bagging

Problem with this approach?

if $z_1 + \dots + z_n$ are independent,
then $\text{var}(\bar{z}) = \frac{\sigma^2}{n}$ decreases ($\downarrow$)

since all observations come from the same dataset, they will be correlated

$$\hookrightarrow \text{var}(\bar{z}) = \frac{\sigma^2}{n} + \frac{1}{2} \text{cov}(z_1, z_2)$$

$\hookrightarrow$ if tree models are correlated, reduction in variance from averaging is limited

Random forest

Random forests provide an improvement over bagged trees by decorrelating trees.

- Each time a split in a tree is considered, a **random sample of m predictors** is chosen as candidates from the full set of p predictors.
- Split is only allowed to use those m predictors.
- A new sample of m predictors is taken at each split.
- Typically we choose $m \approx \sqrt{p}$.
- This process of decorrelating the trees has the effect of reducing the variance.

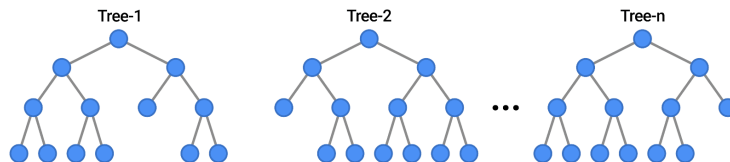
subset of p

↑
that makes
the trees
more diverse

- Bagging: many similar trees
- Random forest: many more **diverse** trees

→ reduces correlation

EXAMPLES



Implementation

See R script: random_forest.R